HW

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HW

Homework 3: Attention Mechanisms

BAC ML

Topic: Attention in Transformer

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Instructions. Show all work and intermediate computations for full credit. Unless otherwise stated, assume all matrices are initialized as given and *do not* apply any learned biases. Write final matrix/vector answers in clearly labeled form. Partial credit is awarded for correct reasoning even with arithmetic errors.

Reference Formulas

Throughout this assignment we use the **column-input convention**: each token embedding is a *column* of $X \in \mathbb{R}^{d_{\text{model}} \times n}$ (d_{model} rows, n token columns). The projection matrices act from the left via their transposes:

$$Q = (W^Q)^\top X, \qquad K = (W^K)^\top X, \qquad V = (W^V)^\top X,$$

where $W^Q, W^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ and $W^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$, giving $Q, K \in \mathbb{R}^{d_k \times n}$ and $V \in \mathbb{R}^{d_v \times n}$.

The scaled dot-product self-attention output is:

$$\text{Attn}(Q, K, V) = V \cdot \text{softmax}\left(\frac{K^\top Q}{\sqrt{d_k}}\right),$$

where softmax is applied *column-wise*: each column of $K^\top Q \in \mathbb{R}^{n \times n}$ is a vector of scores over all n keys for one query, and is normalised independently.

Problems

Problem 1. [10 pts] What Is Attention? A Conceptual Warm-Up.
Consider the following sentence and its Telugu translation:

> **English:** *"Please pay attention when learning about attention or you will get a strike!"*
>
> **Telugu (transliteration):**
> *"Dayacei ceyyam purtci nérvakuntippaide éradéni nahucundi, lékapoté miku strike vastundi!"*
>
> (If viewing on a Unicode-capable device, the original script reads: దయచేసి శ్రద్ధ పూర్తి నేర్వకుంటిప్పైడే ఎరడేని నహుచుండి, లేకపోతే మీకు స్ట్రైక్ వస్తుంది!)

When a human translator reads the Telugu sentence and produces the English word by word, they do not treat every source token equally at every step—some tokens "draw attention" more than others depending on what is currently being generated. Notice also how the threat of a "strike" at the end of the sentence might change which words feel most important! This is the core intuition behind the transformer attention mechanism.

(a) **The analogy.** Using the translation task above as your motivating example, explain in 3–5 sentences what the attention mechanism is doing conceptually. Your answer should address *why* not all tokens should contribute equally to a given output position, and *how* the model decides which tokens are most relevant.

(b) **The Query.** In the translation analogy, the *query* represents the current output position being generated. In one or two sentences, explain what a query "asks for" and why projecting the input into a dedicated query space is preferable to using the raw embedding directly.

(c) **The Key.** Explain what the *key* of a token represents in the same analogy. How does the dot product between a query and a key capture relevance?

(d) **The Value.** Once the model has assigned attention weights (via softmax), it aggregates the *values* of the attended tokens. Explain in one or two sentences what the value represents and why it is useful to have a separate value projection rather than re-using the key.

(e) **Putting it together.** The word "attention" appears *twice* in the English sentence (once as the English content word "attention," once as a transliterated loanword in the Telugu source). If a transformer were translating the Telugu sentence token by token, would you expect the two occurrences to produce *similar* or *different* attention weight patterns over the Telugu source tokens? *Justify* your answer in 2–3 sentences.

1a. The attention mechanism figures out how each word is relevant to the specific word being translated. It determines which words should be prioritized, like aTenshan over the unrelated words so that not all tokens contribute equally to a given output position. The model decides which tokens are most relevant by the Query and Key vectors in which their dot products will represent how relevant each word is to the Query.

1b. A query "asks for" which tokens contain the information that it needs to generate the next. Projecting the input into a dedicated query space is preferable to using the raw embedding directly because it allows the model to get rid of the unnecessary details and essentially filters the input so that it can specifically look for what it needs, providing for better computations and more accurate results.

1c. The key of a token represents the identity of the Telegu token, describing what kind of information it entails. The dot product between a query and a key captures relevance because it shows how relevant the query is to the information the key can provide, demonstrating a higher or lower attention weight for that token depending on a higher or lower dot product.

1d. The value of a token represents the information the token will actually contribute to the final representation. It's useful to have a separate value projection rather than re-using the key because it splits up the two processes, preventing things from getting mixed up or confusing, allowing for greater flexibility.

1e. If a transformer were translating the Telegu sentence token by token, I would expect the two occurrences to produce different attention weight patterns because by using the context, it would be able to identify that the two words are different even though they are technically the same word. Since one is "to pay attention" and the other is learning about attention, the model will be able to distinguish between their two roles in the sentence.

attention weight patterns over the Telugu source tokens? *Justify* your answer in 2–3 sentences.

Problem 2. [10 pts] Hand-Computing Scaled Dot-Product Attention.
Let $n = 2$, $d_k = d_v = 2$. You are given (each column is one token):

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad K = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

(a) Compute the raw attention score matrix $S = K^\top Q$. State its dimension and explain what the entry S_{ij} represents.

(b) Scale S by $\sqrt{d_k}$ to obtain $\tilde{S}$.

(c) Apply softmax *column-wise* to $\tilde{S}$ to obtain the attention weight matrix A. (You may leave answers as exact fractions involving e, or approximate to 4 decimal places.)

(d) Compute the attention output $Z = VA$. State its dimension.

(e) Interpret the first column of A: which token does query 1 attend to more strongly?

(a) $K^T = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$S = K^T Q = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ $n \times n$

The (i,j) entry represents how much token i attends to token j

(b) $\tilde{S} = \frac{S}{\sqrt{2}}$

$\tilde{S} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

(c) $e^{\frac{1}{\sqrt{2}}} = 1.2246$ or $e^{\frac{1}{\sqrt{2}}}/1 = 1.224$ $e^0 = 1$ $e^{\frac{1}{\sqrt{2}}}/e^0 = 1.224$

$A = \begin{bmatrix} 1 & 0.45 \\ 0 & 0.55 \end{bmatrix}$

(d) $Z = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0.45 \\ 0 & 0.55 \end{bmatrix} = \begin{bmatrix} 2 & 0.9 \\ 0 & 1.65 \end{bmatrix}$

$n \times n$ and Token 2 is attended to equally 0.45 times

(e) Token 1, and Token 2 is attended to equally 0.45 times

(b) $\tilde{S} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ $\text{softmax}(\tilde{S})_{1,1} = \frac{e^{\frac{1}{\sqrt{2}}}}{e^{\frac{1}{\sqrt{2}}} + e^0} = 0.55$ $\frac{e^0}{e^{\frac{1}{\sqrt{2}}} + e^0} = 0.45$

(c) $A = \begin{bmatrix} 1 & 0.45 \\ 0 & 0.55 \end{bmatrix}$ $\text{softmax}(\tilde{S})_{2,1} = \frac{e^{\frac{1}{\sqrt{2}}}}{e^{\frac{1}{\sqrt{2}}} + e^0} = 0.55$ $\frac{e^0}{e^{\frac{1}{\sqrt{2}}} + e^0} = 0.45$

Token 1 attends 45% to itself and 55% to token 2, so mostly attends to itself, but also takes some information from token 2

(d) $Z = VA = \begin{bmatrix} 2 & 0.9 \\ 0 & 1.65 \end{bmatrix}$ $n \times n$

Problem 4. [8 pts] Dimensionality Drill.
A transformer model has the following hyperparameters:

$$d_{\text{model}} = 512, \quad d_k = d_v = 64, \quad n = 6 \text{ tokens}.$$

(a) Under the column-input convention, state the dimensions of X, W^Q, and $Q = (W^Q)^\top X$.

(b) State the dimension of the score matrix $S = K^\top Q$.

(c) State the dimension of the attention weight matrix $A = \text{softmax}(\tilde{S}/\sqrt{d_k})$, where softmax is applied column-wise.

(d) State the dimension of the attention output $Z = VA$. Why must its first dimension equal d_v?

Problem 5. [8 pts] Parameter Count.
A transformer encoder has the following configuration:

$$d_{\text{model}} = 256, \quad h = 4 \text{ heads}, \quad d_k = d_v = 64, \quad L = 6 \text{ layers}.$$

(a) With $W^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, how many parameters are in W^Q, W^K, W^V for a single head? How many total across all $h = 4$ heads?

(b) Including the output projection $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$, what is the total parameter count for the attention sub-layer in one layer?

(c) How many total attention parameters are there across all $L = 6$ layers?

(d) Suppose you double d_{model} to 512 (keeping $h = 4$, so $d_k = d_v = 128$). By what factor does the attention parameter count change?

(a) X: $d_{model} \times n$, W^Q: $d_{model} \times d_k$, Q: $(d_k \times n)$

(b) S: $n \times n$

(c) A: $n \times n$

(d) Z: $d_v \times n$, A: $n \times n$

$Z = VA$ $n \times n$

you from the fact that V depends on attention. Z is a weighted sum of the value vectors, and each vector contributes a column of V to the output, and that value vector, d_v tells how many dims of V.

(a) W^Q: $256 \times 64 = 16384$ parameters

W^K: $256 \times 64 = 16384$

W^V: $256 \times 64 = 16384$

Total: $3 \times 16384 = 49152$ parameters

So $3 \times 16384 \times 4 = 196608$ parameters.

(b) W^O: $256 \times 256 = 65536$ parameters

Total: $196608 + 65536 = 262144$ parameters

(c) $262144 \times 6 = 1572864$ parameters

(d) d_{model} to 512: $512 \times 256 = 131072$

Problem 6. [10 pts] Multi-Head Attention — Concepts & Intuition.

> **Background: What Is Multi-Head Attention?**
>
> So far in this course we have studied *single-head* self-attention, where one set of learned projections (W^Q, W^K, W^V) produces a single set of queries, keys, and values. The output is one weighted mixture of value vectors, shaped by a single attention distribution.
>
> **Multi-head attention (MHA)** runs h independent attention "heads" in parallel. Each head i has its own projections (W_i^Q, W_i^K, W_i^V) and computes:
>
> $$\text{head}_i = \text{Attn}\big((W_i^Q)^\top X, (W_i^K)^\top X, (W_i^V)^\top X\big).$$
>
> The outputs of all h heads are *concatenated* along the feature dimension and projected back to d_{model} via an output matrix W^O:
>
> $$\text{MHA}(X) = W^O \begin{bmatrix} \text{head}_1 \\ \vdots \\ \text{head}_h \end{bmatrix}.$$
>
> The standard design choice sets $d_k = d_v = d_{\text{model}}/h$, so each head works in a lower-dimensional subspace. This keeps the total parameter count comparable to a single full-dimensional attention head while allowing the model to attend across multiple *independent representation subspaces simultaneously*.
>
> Empirically, different heads specialise: one head might capture syntactic subject–verb dependencies, another coreference (which noun a pronoun refers to), and another positional proximity. A single head must average all these signals into one distribution and one output—inevitably losing information that competing patterns cannot share.

(a) **Benefits of multi-head attention.** In 4–6 sentences, explain why multi-head attention is preferred over single-head attention in practice. Address at least two distinct advantages, drawing on the intuition in the background above. (*Hint: consider what a single head sacrifices when it must encode multiple types of relationships simultaneously.*)

(b) **Head specialisation and redundancy.** Suppose you train a transformer with $h = 8$ heads on a translation task and, after inspecting the learned attention patterns, you discover that 6 of the 8 heads have converged to nearly identical attention distributions.

6a. (i) Multi-head attention is preferred because it allows the different heads to specialize. A single head, on the other hand, would have to average all these signals into one distribution and one output, losing information. Further, they can run these different versions at the same time since they're independent from each other. With more aspects being taken into account and each head specializing in these, it will lead to more accurate results.

6b. (i) It could mean the model is too big and doesn't need all of that space and 8 heads is too much for a simple task. But, it could also be that the model found a solution that was good enough and didn't see the need to utilize the other heads.

6b. (ii) This might encourage greater diversity because since it's random, it will prevent some heads from becoming too similar to each other and lead to more different results. The model will be forced to diversify and focus on other features to "cover up" for the missing ones, preventing it from relying on a similar pattern.

(i) What does this suggest about how the model is using its representational capacity? Is this necessarily a problem, or could there be a benign explanation?

(ii) Researchers have proposed *attention dropout*—randomly zeroing entire heads during training—as a regularisation technique. In 2–3 sentences, explain how this might encourage greater diversity across heads and reduce redundancy.

Good luck! Remember: understanding the shapes is half the battle.